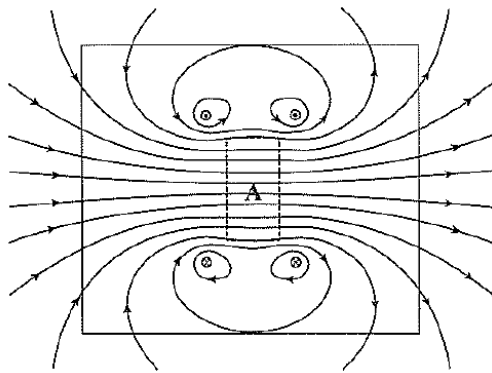


	(For the same E), balance length decreases.	A1
7a	The Helmholtz coil is made up of two flat circular coils, with a fairly uniform field strength in their central axis, similar to that of a solenoid. The magnetic field pattern is shown below.  Marking points: 1. Direction of field lines indicated correctly on every line 2. Overall pattern, resembling that of a solenoid, extending over the whole area of the board. 3. In region A, field is approximately uniform, with lines of equal spacing.	B1 B1 B1
7bi	$B = \frac{\mu_0 I_X}{2\pi d}$ $= \frac{4\pi 10^{-7} \times 90}{2\pi (0.050)}$ $= 3.6 \times 10^{-4} \text{ T}$	C1 A1
7bii	Force by wire X on wire Y = weight of wire Y $BI_Y L = mg$ $\frac{m}{L} = \frac{BI_Y}{g}$ $= \frac{3.6 \times 10^{-4} \times 60}{9.81}$ $= 2.2 \times 10^{-3} \text{ kg m}^{-1}$	M1 A1
7c	Electric force is independent of the speed whereas for magnetic force, it is proportional to the speed. or Electric force is along the field direction, whereas for magnetic force, it is perpendicular to the field direction.	B1
7di	Into the plane of paper	A1
7dii	Electromagnetic force provides the centripetal force for the particle. $F_B = \frac{mv^2}{r}$ $Bqv = \frac{mv^2}{r}$ $Bqr = mv$ Hence momentum is proportional to the radius.	B1 M1

	$\frac{\text{final momentum of particle}}{\text{initial momentum of particle}} = \frac{5.7}{7.4} = 0.77$	A1
7ei	Faraday's Law of electromagnetic induction states that the <u>e.m.f. induced</u> in a conductor is <u>proportional to the rate of change of magnetic flux linkage</u> (or rate of cutting of the flux).	A1
7eii	The <u>changing/alternating</u> current in coil A produces a <u>changing/alternating magnetic flux density (B)</u> (since magnetic flux density is directly proportional to the current). Hence, <u>coil B experiences a changing magnetic flux linkage</u>. By Faraday's law, an e.m.f. is induced in coil B.	B1 B1 A0
7eiii	t = 5.0 ms, 15.0 ms or 25.0 ms	A1
7eiv	Since the <u>gradient of the tangent of the graph</u> at this point is <u>the largest</u> , it corresponds to the <u>greatest rate of change of current/magnetic flux density/magnetic flux linkage</u> . Hence, induced e.m.f. is a maximum.	A1
7ev	From t = 10 ms to 15 ms, <u>current in coil A is decreasing</u>. The <u>magnetic flux linkage through coil B is decreasing</u>. According to Lenz's law, the induced e.m.f. will be directed such that the current it causes in the coil B will <u>produce a flux to oppose this decrease in flux in coil B</u>. Hence, the current in coil B flows in the <u>same direction</u> as coil A.	M1 M1 A1
8ai1	Gain in KE = loss in EPE $KE_{\text{final}} - KE_{\text{initial}} = q\Delta V$ (as electron starts from rest, $KE_{\text{initial}} = 0$) $KE_{\text{final}} = 140 \text{ eV}$	A1
8ai2	$E_K = p^2 / 2m$ $p = (2m E_K)^{1/2}$ $= (2 \times 9.11 \times 10^{-31} \times 140 \times 1.6 \times 10^{-19})^{1/2}$ $= 6.39 \times 10^{-24} \text{ kg m s}^{-1}$ Alternative method: $\frac{1}{2} mv^2 = q \Delta V$ $\frac{1}{2} (9.11 \times 10^{-31}) v^2 = (140) \times 1.60 \times 10^{-19}$ $v = 7.013 \times 10^6 \text{ m s}^{-1}$ $p = mv = (9.11 \times 10^{-31}) (7.013 \times 10^6)$ $= 6.39 \times 10^{-24} \text{ kg m s}^{-1}$	C1 A1
8aii1	Electrical potential energy to kinetic energy when accelerated by potential difference. Kinetic energy to light energy upon colliding with fluorescent screen.	B1 B1
8aii2	The <u>electrons</u> must exhibit a smaller <u>de Broglie's wavelength λ</u> ($d \sin \theta = n\lambda$). According to de Broglie's equation ($\lambda = h / mv$), in order to have a smaller λ, the momentum / velocity of the electron must be larger. This is done by increasing the potential difference between the electrodes.	M1 M1 A1

Φ_{hi}	$(-5.15, +5.00)$	10^{-19}	0.15	10^{-19}	1	
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